

Hierarchical Structure of Adaptation Burden Under Drift

Scaffold

Abstract

Adaptation burden under drift is not determined by a single scalar notion of memory quality. The empirical object in this paper is the minimum realizable re-learning burden

$$R^*(m, \pi) := \min_w R(m, \pi, w),$$

defined over protocol-window choices after drift. The main empirical result is a hierarchical decomposition. Across architectural families, effective spectral rank separates degradation regimes. Within a recurrent local regime, activation velocity Δ_v predicts burden strongly even when effective rank still varies substantially. The pooled data therefore support a two-level structure: rank is a family-level regime coordinate, while Δ_v is a within-regime scale coordinate. This decomposition makes clear why static endpoint metrics are insufficient: models with similar task performance can occupy different burden regimes and require different post-drift re-learning effort.

1 Introduction

Temporal representation quality under drift is not exhausted by endpoint task accuracy. A model can retain acceptable pre-drift performance and still require substantially different post-drift re-learning effort from another model with similar static accuracy. The empirical question is therefore not only whether a representation is accurate, but how costly it is to re-tune once the signal changes.

The operational object in this paper is the minimum realizable adaptation burden over protocol-window choices,

$$R^*(m, \pi) := \min_w R(m, \pi, w).$$

Here m denotes the trained model family and π denotes the adaptation protocol. This object connects the empirical program to the temporal-validity functional $V(\phi, \tau)$: inferential utility curves over horizon identify how useful present representations remain for future targets, while R^* measures the downstream cost of restoring utility after drift.

The central empirical claim is hierarchical rather than universal. Effective spectral rank separates degradation regimes across architectural families. Within a recurrent local regime, activation velocity Δ_v predicts burden more strongly than effective rank, even though effective rank still varies over a wide

range. Those two facts imply that family choice and within-family scale choice move burden through distinct empirical channels.

That hierarchy matters because the literature often compresses stability, plasticity, representational geometry, and memory into a single tradeoff. The results below instead identify two observational levels. Family-level geometry determines which burden regime a model enters. Within at least one recurrent family, dynamic reactivity determines where the model sits inside that regime.

2 Results

Empirical objects

The inferential object is the utility curve $U(\tau)$, defined by linear access to future-relevant targets from present representations across horizons τ . The summary observables are inferential half-life, utility area, and the adaptation burden target

$$R^*(m, \pi) = \min_w R(m, \pi, w),$$

the minimum post-drift loss area over adaptation-window choices for protocol π . This definition lets the burden comparison absorb the non-monotone window effects already present in recurrent multiscale adaptation.

Family-level regime separation

The first empirical regularity is a family-level ordering of burden regimes. Under the fast four-seed benchmark with the full window grid $w \in \{1, 16, 64\}$, the abrupt-drift mean burdens are approximately 2.1, 2.2, 3.2, and 12.2 for feedforward local, feedforward multiscale, recurrent local, and recurrent multiscale variants respectively. Under gradual drift the means are approximately 2.8, 2.7, 9.3, and 22.3 in the same order. The qualitative hierarchy therefore persists across the two tested drift types: feedforward variants occupy the low-burden regime, recurrent local lies intermediate, and recurrent multiscale is the highest-burden regime.

Effective spectral rank is the strongest pooled regime separator in the current dataset. On the pooled fast benchmark ($n = 32$ cases), $\text{Spearman}(\text{rank}, R^*) \approx 0.74$, whereas $\text{Spearman}(\Delta_v, R^*) \approx 0.02$. A pooled additive linear model using z-scored rank and Δ_v reaches $R^2 \approx 0.38$, and adding the interaction term $\text{rank} \times \Delta_v$ improves fit only marginally ($\Delta R^2 \approx 0.005$). At the pooled level, effective rank therefore functions as a family-level regime coordinate, while Δ_v does not act as a universal burden predictor.

Within-family scale structure in recurrent local models

The second empirical regularity appears inside a fixed family. In the recurrent local variant, a targeted capacity sweep over hidden $\in \{32, 48, 64, 96\}$ and depth $\in \{2, 3, 4\}$ under abrupt drift yields $\text{Spearman}(\Delta_v, R^*) \approx 0.71$, while $\text{Spearman}(\text{rank}, R^*) \approx 0.36$. This is not a low-variance artifact: effective rank still spans approximately 17.6 to 65.5 across the same sweep, yet its Pearson correlation with R^* is only moderate (≈ 0.43). Within this recurrent local regime, dynamic reactivity is therefore the stronger burden coordinate.

The distinction between those two levels is the main empirical result. Rank varies enough within the recurrent local family to rule out a trivial variance explanation, but it still does not dominate burden there. The data support a hierarchical structure: family choice sets the degradation regime, while within-family scale variation shifts burden through the activation-velocity channel.

Cross-drift prognosis remains sample-limited

The prognosis layer is feasible but not yet closed. On the fast four-seed run, a gradual-to-abrupt ridge forecaster based on routing alignment, deep incremental utility area, inferential half-life difference, and peak-gap information reaches $\rho \approx 0.66$ but negative R^2 and does not dominate the strongest single-feature baseline. Effective rank alone gives the best rank correlation in that setting ($\rho \approx 0.89$), while Δ_v yields weaker pooled generalization. The current evidence therefore supports prognosis as a viable layer, but not yet as a closed predictive theorem for burden across drift types.

Architecture-moderated rank effects remain open

Exploratory diagnostics suggest that the semantics of spectral rank may depend on architecture. In reduced-budget within-variant tests, the recurrent local slice shows a strong positive relation between Δ_v and burden, whereas feedforward local behavior does not follow the same pattern and can even display sign reversals for rank in very small samples. A pooled moderation fit with an architecture indicator and rank $\times$ architecture interaction improves fit modestly in a reduced dataset, but the coefficients remain unstable at that sample size. The supported conclusion is therefore hierarchical rather than fully mechanistic: rank separates families, and Δ_v controls burden inside the recurrent local regime.

3 Limitations

The architecture-moderated semantics of spectral rank are not yet settled. The current feedforward within-family diagnostics are short-run and low-sample, so they do not support a stable sign claim for rank inside feedforward regimes. Likewise, the prognosis layer remains sample-limited: the current cross-drift forecaster ranks cases nontrivially, but does not yet dominate strong single-feature baselines.

The strongest within-family result is presently confined to the recurrent local regime. Extending the same scale-level monotonicity claims to recurrent multiscale or feedforward families requires additional controlled sweeps.

Those limitations do not affect the main supported structure. The family-level role of effective rank and the within-family role of Δ_v in recurrent local models are empirically distinct and reproducible in the tested settings.

4 Conclusion

Adaptation burden under drift is not organized by a single memory coordinate. The supported empirical structure is hierarchical. Across families, effective spectral rank identifies degradation regimes. Within a recurrent local regime,

activation velocity controls burden more strongly than rank even when rank still varies broadly. That separation explains why static endpoint metrics are blind to temporal adaptation quality: regime selection and within-regime scale control move burden through different channels.

The next theorem-level empirical step is not to enlarge the current claim, but to stabilize the open mechanism. Architecture-moderated rank effects and broader within-family burden laws remain open problems built on top of the hierarchical structure established here.